IEEE 802.11 Frequency Offset Estimation and Correction

Sourour 2004 paper reference

IEEE 802.11

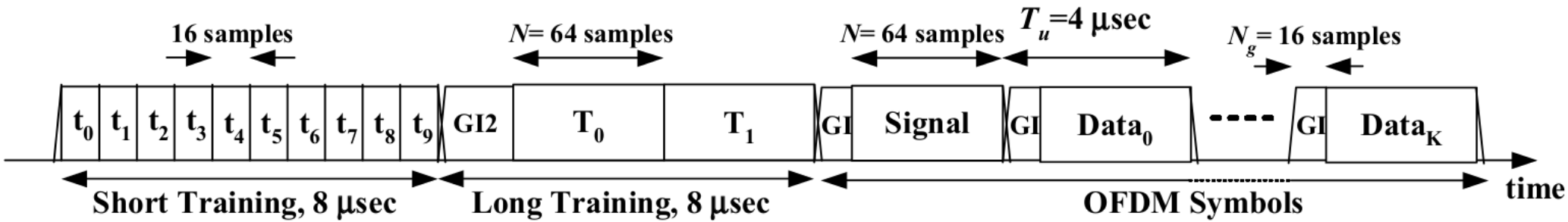


Figure 1, IEEE 802.11a frame structure

	Short	Long
Duration	8 µs	8 µs
Parts	10 ($t_0, \ldots, t_9$)	2 (T_0, T_1)
Samples/part	16	64
Guard Interval	—	GI2 (32 samples)
Applications	Frame sync · AGC · Antenna sel. · Coarse CFO	Channel est. · Fine CFO

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Pilot Subcarriers

In each OFDM symbol, four of the subcarriers are dedicated to pilot signals in order to make the coherent detection robust against frequency offsets and phase noise. These pilot signals shall be put in **subcarriers –21, –7, 7, and 21**. The pilots shall be BPSK modulated by a **pseudo-binary sequence** to prevent the generation of spectral lines. 18.3.5.9.

CFO Rotation Per Sample

α represents the **rotation per sample** due to CFO

$$\alpha = 2\pi\Delta_f T_s$$

where Δ_f is the frequency error between the received carrier and the LO.
To model the effect of CFO, sample n of the received frame is multiplied by $e^{jn\alpha}$.

Short Training (Coarse)

With a short training consisting of 10 repeated parts with 16 complex-value samples each part (IEEE 802.11 PHY) , lets denote de ST complex samples of the last 5 parts as S_m, where $m = 0, 1, \ldots 79$.

$$P = \sum_{m=0}^{63} S_m^* S_{m+16}$$

has an angle of:

$$\angle P = 16\alpha_{ST}$$

Short Training (Coarse)

Estimating α_{ST}:

$$\boxed{\hat{\alpha}_{ST} = \frac{1}{16}\angle P}$$

Due to the restriction $-\pi < \angle(.) \leq \pi$:

$$-\pi < 16\alpha \leq \pi$$

$$\boxed{-\frac{\pi}{16} < \alpha \leq \frac{\pi}{16}}$$

Ambiguity over subcarrier spacing:

$$-\frac{\pi}{16} < 2\pi\Delta_f T_s \leq \frac{\pi}{16}$$

$$-\frac{\pi}{16} < 2\pi(\frac{\epsilon}{NT_s})T_s \leq \frac{\pi}{16}$$

With $N = 64$ (IEEE 802.11 PHY):

$$-\frac{\pi}{16} < 2\pi\frac{\epsilon}{64} \leq \frac{\pi}{16}$$

$$\boxed{-2 < \epsilon \leq 2}$$

Long Training (Fine)

Calculating as same as for the short training:

$$P = \sum_{m=0}^{63} S_m^* S_{m+64}$$

$$\angle P = 64\alpha_{LT}$$

$$\boxed{\hat{\alpha}_{LT} = \frac{1}{64}\angle P}$$

$$\boxed{-\frac{\pi}{64} < \alpha \leq \frac{\pi}{64}}$$

$$-\frac{\pi}{64} < 2\pi\frac{\epsilon}{64} \leq \frac{\pi}{64}$$

$$\boxed{-\frac{1}{2} < \epsilon \leq \frac{1}{2}}$$

Estimation and Correction Process

1. Coarse estimation α_{ST}.
2. Coarse correction: The 128 samples of LT are corrected as $S_m = S_m e^{-jm\hat{\alpha}_{ST}}, m = 0, 1 \ldots 127.$
3. Fine estimation α_{LT}.
4. Fine correction: The samples are corrected one more time as $S_m = S_m e^{-jm\hat{\alpha}_{LT}}, m = 0, 1 \ldots 127.$

The final phase estimation $\hat{\alpha} = \hat{\alpha}_{ST} + \hat{\alpha}_{LT}$ is used for carrier frequency correction for the remaining part of the frame after LT.

Channel Estimation

Channel estimation

The parts T_0 and T_1 are known by the receptor, and are used for channel estimation. We denote the channel gain estimates as H_k, $k = -26 \dots 26$. The Pilot Symbols are given in the subcarrier positions $k = \{-21, -7, 7, 21\}$. We represent the **channel gains at the Pilot Symbols locations** as:

$$Q_{-2} = H_{-21}$$

$$Q_{-1} = H_{-7}$$

$$Q_{1} = H_{7}$$

$$Q_{2} = H_{21}$$

At the receiver, the FFT output is denoted as $X_{l,k}$, where k is the subcarrier number, $k = -26 \dots 26$, and l is the OFDM Symbol numbre starting from $l = 1$. For simple notation, **Pilot Symbols at the l'th OFDM symbol** are referred as:

$$P_{l,-2} = X_{l,-21}$$

$$P_{l,-1} = X_{l,-7}$$

$$P_{l,1} = X_{l,7}$$

$$P_{l,2} = X_{l,21}$$

Residual CFO and SFO

In the presence of **residual frequency offset and sampling frequency offset** the FFT outputs, including the Pilot Symbols, will incur phase rotation. The Pilot Symbols $P_{l,i}$ and the corresponding channel gains $Q_i, i = -2, -1, 1, 2$, are used to estimate this offset. It is shown that due to frequency offset, each FFT output $X_{l,k}$ incurs extra phase rotation $\varphi_{l,k}$ given by:

$$\varphi_{l,k} = 2\pi l \left(\epsilon_r T_u f_c - \frac{N + N_g}{N} \epsilon k \right)$$

where

- $N = 64$ is the FFT size
- $N_g = 16$ is the guard samples length
- $T_u = 4$ usec is the OFDM Symbol duration
- $\epsilon_r = \epsilon - \hat{\epsilon}_0$ is the residual carrier frequency offset after LT correction, which is nonzero if $\hat{\alpha} \neq \alpha$

The **first part in the right hand side** of the equation is due to the residual carrier frequency offset not corrected after LT correction. This term is equal for all subcarriers of the same OFDM Symbol.
The **second part** is due to the continuous drift in the sampled waveform of the OFDM Symbol due to sampling offset. This phase depends linearly on the subcarrier number.

Steps to correct rotations

1)Correct Sampling Frequency Offset

2)Correct Residual Carrier Frequency Offset

3)Fine Estimation in Frequency Offset

1. Correct Sampling Frequency Offset

we first compensate for the second term in the right hand side of the equation. Using the available frequency offset estimate$\hat{\epsilon}_{l-1}$ the l'th OFDM Symbol FFT output is updated as:

$$X_{l,k} = X_{l,k}\, exp\left(2\pi l\, \frac{N+N_g}{N}\, \hat{\epsilon}_{l-1} k\right) \quad , k = -26 \ldots 26$$

Note that this correction includes the Pilot Symbols. When processing the Signal field (l=1) the value of $\hat{\epsilon}_0$ is the one generated from the LT processing. As more OFDM Symbols are received the frequency offset estimate $\hat{\epsilon}_l$ improves.

2. Correct Residual Carrier Frequency Offset

First term in the right hand side of the equation is due to the accumulated phase rotation, at the l'th OFDM Symbols, when $\hat{\alpha} \neq \alpha$. A good estimate of it is given by:

$$\hat{\beta}_l = \angle \left(\sum_{i=-2,-1,1,2} P_{l,i} \, Q_i^* \right)$$

This is the correlation between the Pilot Symbols, corrected in the previous step, and the channel gain at the corresponding subcarrier. Again, see Angle of an autocorrelation process if it is not clear.

Now the output of the previous step is corrected one more time as follows:

$$X_{l,k} = X_{l,k} \, e^{-\hat{\beta}_l} \quad , k = -26 \ldots 26, \quad k \neq -21, -7, 7, 21$$

The Pilot Symbols are not corrected since they will be used in the next step for finer estimation of the frequency offset estimate $\hat{\epsilon}_l$.

3. Fine Estimation of Frequency Offset

The goal is to improve $\hat{\epsilon}_l$ to be used in step 1 for the following OFDM Symbol.

We first calculate the variable W_l as follows:

$$W_l = \sum_{i=-2,-1,1,2} P_{l,i}\, P^*_{l-1,i}$$

We notice that $\angle(W_l)$ is an estimate of the phase increment $2\pi\epsilon_r T_u f_c$ over one OFDM symbol, due to residual carrier frequency offset.

3. Fine Estimation of Frequency Offset

To improve the accuracy of estimation we could continuously accumulate W_l every new OFDM symbol. However, **continuous addition would be problematic in a fixed-point hardware implementation**. As a trade off we add groups of 4 successive values and exponentially filter the sum.

Addition:

$$V_l = \sum_{m=l-3}^{l} W_m$$

Filtering:

$$U_l = \rho V_l + (1-\rho) U_{l-4}$$

The parameter ρ is taken to be 1/32.

3. Fine Estimation of Frequency Offset

Every 4'th OFDM Symbol a filtered value U_l is generated, and a corresponding new estimates ϵ_r and $\hat{\epsilon}_l$ are updated as follows:

$$\hat{\epsilon}_r = \angle(U_l)/(2\pi T_u f_c) \quad , l = 4, 8, 12, 16, \ldots$$

$$\hat{\epsilon}_{l+1} = \hat{\epsilon}_{l+2} = \hat{\epsilon}_{l+3} = \hat{\epsilon}_l = \hat{\epsilon}_0 + \hat{\epsilon}_r \quad , l = 4, 8, 12, 16, \ldots$$

CFO Estimation and Correction in IEEE 802.11

